SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence and Data Science ASSESSMENT TOOL 3 – Class Test

Course Code:	DSA03	**Course Title:**	Natural Language Processing
CO Assessed:	CO2 – Manipulation	**Assessment:**	Assessment Tool 3 – Class Test
Weightage:	20% of CIA		
CO5 Statement:	Design inflectional, derivational, finite-state parsing, stemming algorithms and for analyse morphological methods. (BL-3)		
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Section / Batch:	CSE-AIML	**Date:**	29/07/26

Code of Conduct

*Iam **A.Shiva Shankar Reddy/ 192425174** certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.* Signature of the Student: ________A.Shiva Shankar Reddy_________________

Instructions

- Read all questions carefully before attempting the answers.
- Answer all five questions. Each question carries equal marks.
- Show all intermediate steps, calculations, probability computations, tagging decisions, and justifications wherever applicable.
- Duration: 30 minutes.

Section / Topic	Focus	Q Nos
N-gram Models and Word Counting for Language Modeling	N-gram Construction, Word Frequency Analysis, Probability Estimation	1

Unsmoothed N-grams and Smoothing Techniques in Syntax Analysis	Unsmoothed N-grams, Backoff, Deleted Interpolation	2
Entropy and Language Uncertainty in NLP Systems	Information Theory, Entropy Measurement, Language Prediction	3
English Word Classes and Tagsets for Linguistic Analysis	Word Categories, POS Categories, English Tagsets	4
Part-of-Speech Tagging Techniques and Applications	POS Tagging, Rule-Based Tagging, Stochastic Tagging, Transformation-Based Tagging	5

Annexure A – Class Test

1. A document processing system encounters the words "talked", "talking", and "talks". Apply inflectional morphology rules and Porter Stemmer steps to derive the normalized root form. Identify the suffixes removed and explain the role of morphological normalization in improving information retrieval accuracy.

2. Given the input words: "relational", "conditional", "happiness", "running", design a hybrid morphological processing pipeline using derivational rules, inflectional analysis, finite-state morphological parsing, and Porter Stemmer. Decompose each word into root and affixes, represent transitions using finite-state parsing logic, and apply stemming steps to obtain the final normalized forms. Analyze how each technique contributes to accurate root extraction.

3. An enterprise-level search engine optimization (SEO) system processes web content containing variations such as "optimize", "optimization", "optimized", and "optimizing". The system must ensure consistent keyword indexing and ranking. Develop a morphology-driven solution integrating inflectional and derivational analysis, finite-state morphological parsing for rule validation, and Porter Stemmer for normalization. Apply the approach to the given inputs and determine how the system achieves improved search relevance and reduced redundancy.

4. A large-scale information retrieval system processes input words: "compute", "computer", "computing", and "computation" to enhance search accuracy.
Design a hybrid morphological pipeline integrating inflectional and derivational analysis, finite-state morphological parsing, and Porter Stemmer.
Decompose each word into root and affixes, represent transformations using state transitions, and apply stemming rules for normalization.
Ensure consistent root extraction across all variations for efficient indexing.
Determine the morphological breakdown, parsing transitions, and final normalized forms for each word.

5.A text mining system processes input words: “decide”, “decision”, “decisive”, and “deciding” to support semantic clustering and analysis.
Construct a morphology-based framework combining derivational rules, inflectional transformations, finite-state parsing, and Porter stemming.
Analyze how suffix variations affect meaning and grammatical structure while maintaining root consistency. Apply the approach to extract roots, validate transformations, and normalize the dataset. Derive the morphological structure, state transitions, and final stem for each input word.

Q1. Inflectional Morphology and Porter Stemmer

Input Words

talked, talking, talks

Step 1: Morphological Decomposition

Word Root Suffix talked talk -ed
talking talk -ing talks talk -s

Step 2: Inflectional Analysis

Word Type

talked Past tense (Inflectional) talking Present participle (Inflectional) talks Third-person singular (Inflectional)

Step 3: Porter Stemmer

- **talked → remove -ed → talk**
- **talking → remove -ing → talk**
- **talks → remove -s → talk**

Final Normalized Form

Word Normalized Root
talked talk talking talk
talks talk

Role of Morphological Normalization

All grammatical forms are mapped to talk, improving search accuracy, reducing duplicate entries, and increasing information retrieval efficiency.

Q2. Hybrid Morphological Processing Pipeline

Input Words

relational, conditional, happiness, running

Step 1: Morphological Decomposition

Word Root Affix

relational relate -ional conditional
condition -al

happiness happy -ness running

run -ing

Step 2: Inflectional / Derivational Analysis

Word

Type relational

Derivational conditional

Derivational happiness

Derivational running

Inflectional

Step 3: Finite-State Parsing

relate → relational

condition → conditional

happy → happiness run →

running

Step 4: Porter Stemmer

Word	**Stem**
relational	**relat**
conditional	**condit**
happiness	**happi**
running	**run**

Final Normalized Forms

Word	**Final Stem**
relational	**relat**
conditional	**condit**
happiness	**happi**
running	**run**

Contribution of Each Technique

- **Derivational analysis: Identifies word formation.**
- **Inflectional analysis: Handles grammatical changes.**
- **Finite-state parsing: Validates transformations.**
- **Porter Stemmer: Produces a common normalized stem.**

Q3. Morphology-Driven SEO Solution

Input Words

optimize, optimization, optimized, optimizing

Step 1: Morphological Decomposition

Word	**Root**	**Affix**
optimize	**optimize**	**—**

Word
optimization optimize -ation optimized

optimize -ed Root Affix

optimizing optimize -ing
Step 2: Classification

Word Type

optimize Base

optimization Derivational optimized

Inflectional optimizing

Inflectional

Step 3: Finite-State Parsing
optimize ├── optimized
├── optimizing
└── optimization
Step 4: Porter Stemmer

Word Final Stem

optimize optim

optimization optim optimized

optim optimizing

optim

Final Normalized Form
All words → optim
Result
The search engine indexes all forms under one keyword, improving search relevance and reducing redundant indexing.

Q4. Hybrid Morphological Pipeline
Input Words
compute, computer, computing, computation
Step 1: Morphological Decomposition

Word Root Affix compute

compute — computer

compute -er computing

Word

compute -ing computation

compute -ation

Step 2: Classification

Type

compute Base computer

Derivational

computing Inflectional

computation Derivational

Step 3: Finite-State Parsing

```
compute ├── computer
├── computing
└── computation
```

Step 4: Porter Stemmer

Word Final Stem

compute comput computer

comput computing

comput computation

comput

Final Normalized Form

All words → comput

Q5. Morphology-Based Framework

Input Words

decide, decision, decisive, deciding

Step 1: Morphological Decomposition

Word Root Affix decide decide

—

decision decide -sion

decisive decide -ive

deciding decide -ing

Step 2: Classification

Word

Word	Type
decide	Base
decision	Derivational
decisive	Derivational
deciding	Inflectional

Step 3: Finite-State Parsing decide

├── decision
├── decisive
└── deciding

Step 4: Porter Stemmer

Word Final Stem
decide decis
decision decis decisive
decis deciding decis

Final Normalized Form
All words → decis Effect of Suffix Variations

- **-sion forms a noun.**
- **-ive forms an adjective.**
- **-ing indicates present participle.**
- **All forms are normalized to the common stem decis for semantic clustering and efficient retrieval.**

Rubrics for Calculation

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Task	20%	Clearly understands the given task/experiment without assistance	Understands task with minor clarification	Partial understanding; requires guidance	Unable to understand the task
Procedure & Execution	25%	Performs procedure accurately and independently with proper technique	Performs with minor errors	Requires guidance; makes procedural mistakes	Unable to execute the procedure
Observation & Result	20%	Records accurate observations and obtains correct results	Minor errors in observation or result	Incomplete or partially correct results	Incorrect or no results

Viva Voce / Conceptual Response	20%	Answers all questions confidently with strong conceptual clarity	Answers most questions with minor gaps	Limited responses; needs prompting	Unable to answer questions
Time Management & Presentation	15%	Completes task on time with neat and clear presentation	Completes with minor delay or presentation issues	Delayed or poorly presented work	Unable to complete on time

Q. No. / Criteria Level	**Understanding of Task**	**Procedure & Execution**	**Observation & Result**	**Viva Voce / Conceptual Response**	**Time Management & Presentation**	**Sub. Total**	**Total %**
1							
2							
3							
4							
5							

Faculty In-charge	**Course Coordinator**	**Program Director**
Signature: _______________	Signature: _______________	Signature: _______________
Date: ___________________	Date: ___________________	Date: ___________________